MIL-C-47273(MI)
9 August 1974
SUPERSEDING
MIS-18132
25 September 1967
MIS-18133
25 September 1967

MILITARY SPECIFICATION
CLEANER, PRINTED CIRCUIT ASSEMBLY

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a special purpose solution used for removing surface contaminants from circuit card assemblies.

1.2 Classification. Printed circuit assembly cleaner covered by this specification shall be of the following types, as specified:

Type I - For usage to remove oils, greases, and fingerprint type contaminants.

Type II - For usage to remove inorganic contaminants such as salt deposits and corrosion products after precleaning with Type I cleaning solution.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Government Property

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:08Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall meet all requirements of this specification.

3.2 General Requirements. All materials used in the manufacturing of this cleaning solution shall be of the best commercial grade available.

3.3 Solvency. This cleaning solutions shall be completely removable when subjected to a running water rinse.

3.4 Chemical composition.

3.4.1 Type I. The Type I cleaning solution shall contain surfactants, wetting agents and organic solvents capable of removing oil, grease, and fingerprint type deposits from circuit card surfaces.

3.4.1.1 Ph. The Type I cleaning solution shall be of a neutral nature and non-corrosive to aluminum. The ph shall range from 6.5 to 7.

3.4.2 Type II. The Type II cleaning solution shall contain surfactants, wetting agents and organic solvents capable of removing inorganic contaminants from circuit card surfaces.

3.4.2.1 Ph. The Type II cleaning solution shall have a ph of 11 to 12. The solution shall incorporate a silicate inhibitor to prevent corrosion of aluminum.

3.5 Physical Properties.

3.5.1 Type I. The specific gravity of the Type I cleaning solution shall be approximately 0.865.

3.5.2 Type II. The specific gravity of the Type II cleaning solution shall be approximately 1.16.

3.6 Toxicity - safety. The cleaner shall contain no carbon tetrachloride nor other highly toxic materials. No solvents shall be used that have a threshold limit of less than 100 parts per million for an eight hour work day (as printed in the latest edition of American Medical Association Archives of Environmental Health). Warnings shall be listed in 5. herein for precautions needed to be observed in the use of this material.

Source: https://assist.dla.mil -- Downloaded: 20260-22T08:08Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISION

4.1 Classification of tests. Inspection and testing of the cleaning solutions shall be classified as follows:

a. Preproduction inspection tests.

b. Quality conformance tests.

4.1.1 Preproduction inspection tests. Preproduction inspection tests shall be performed only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.1.2 Quality conformance tests. Quality conformance tests for acceptance of the cleaning solution shall be as specified below.

4.1.2.1 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.2 Test methods.

4.2.1 Test procedures.

4.2.1.1 Oil removal, type I only.

4.2.1.1.1 Preparation of circuit card. A thin film of commercial grade 20 weight oil shall be applied to the surface of a circuit card and baked in an oven at a temperature of 250 degrees Fahrenheit (F) for a period of 2 hours plus or minus 10 minutes.

4.2.1.1.2 Cleaning. Immersion of the card in the cleaning solution shall remove all traces of the oil in a period of less than 10 minutes.

4.2.1.2 Inorganic Contaminant removal, type I and type II.

4.2.1.2.1 Preparation of circuit card. A printed circuit card shall be exposed to a solution of ammonium carbonate and ammonium hydroxide which results in the formation of blue corrosion products on the copper conductors. A thin film of oil shall be baked on the surface as specified in 4.2.1.1.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T08:08Z
Check the source to verify that this is the current version before use.

4.2.1.2.2 Cleaning. After cleaning with the Type I solution as specified in 4.2.1.1.2, followed by immersion in the Type II cleaning solution for a period of 10 plus or minus 2 minutes, there shall be no residual traces of corrosion product.

5. PREPARATION OF DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing, and marking shall be in accordance with MIL-STD-130.

5.2 Warning notice. The following warning shall be printed on each individual container and each shipping container:

> Keep container closed when not in use. Provide adequate ventilation. Avoid inhalation of vapors and contact with skin.

If the cleaner is flammable, the following shall also be printed:

> Flammable! Keep away from heat, sparks, and open flame.

6. NOTES

6.1 Intended Use. Type I and Type II cleaning solutions are intended to be used for cleaning surface contaminants from circuit card assemblys.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type

c. Whether a preproduction sample is required and, if so, pertinent details.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specifications MIS-18132 and MIS-18133, both dated 25 September 1967.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 6810-AA16

Source: https://assist.dla.mil -- Downloaded: 202610-22T08:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-22T08:08Z
Check the source to verify that this is the currentversion before use.